AHMAD BEYH, PHD

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EMPLOYMENT

Postdoctoral Associate 2023–
Department of Psychiatry, Brain Health Institute, Rutgers University
Mentor: Dr Linden Parkes

Visiting Researcher 2023–2024
Donders Institute for Brain, Cognition and Behaviour, Radboud University
Mentor: Dr Stephanie Forkel

Postdoctoral Fellow 2021–2023
Department of Cell & Developmental Biology, University College London
Mentor: Prof Semir Zeki

Research Worker 2016–2021
Centre for Neuroimaging Sciences, King's College London

EDUCATION

PhD, Neuroimaging 2022
King's College London
Thesis: "The functional anatomy of white matter pathways for visual configuration learning"
Supervisors: Prof Marco Catani, Prof Dominic Ffytche, Dr Flavio Dell'Acqua

MSc, Neuroscience 2015
King's College London
Thesis: "A retinotopic connectivity atlas of the human occipital lobe"
Supervisor: Prof Marco Catani
Graduated with distinction

BA, Psychology 2014
American University of Beirut
Thesis: "An EEG study of self-control, risk-taking, and the role of the prefrontal cortex"
Supervisors: Prof Charles Harb, Prof Arne Dietrich

PUBLICATIONS

* Contributed equally.

Peer-Reviewed Articles

17. Kim JZ, Betzel R, **Beyh A**, … Parkes L (2025). Inferring macroscopic intrinsic neural timescales using optimal control theory. *Nature Communications*.
16. Lim L, Radua J, **Beyh A** (2025). Structural connectivity abnormalities in suicidal thoughts and behaviours: a meta-analysis. *Translational Psychiatry*.
15. Mehra C, **Beyh A**, … O'Muircheartaigh J (2025). Zero-phase-delay synchrony between interacting neural populations: implications for functional connectivity-derived biomarkers. *Imaging Neuroscience*.
14. **Beyh A**, Rasche S, Leff A, Ffytche D, Zeki S (2024). A clinico-anatomical dissection of the magnocellular and parvocellular pathways in a patient with the Riddoch syndrome. *Brain Structure & Function*.

13. **Beyh A**, Howells H, Giampiccolo D, … Vergani F (2024). Connectivity defines the distinctive anatomy and function of the hand-knob area. *Brain Communications*.

12. Rasche S* & **Beyh A***, Paolini M, Zeki S (2024). Neural correlates of the experience of ugliness. *European Journal of Neuroscience*.

11. Zeki S, Hale ZF, **Beyh A**, Rasche S (2024). Perceptual axioms are irreconcilable with Euclidean geometry. *European Journal of Neuroscience*.

10. **Beyh A**, Rasche S, Leff A, Ffytche D, Zeki S (2023). Neural patterns of conscious visual awareness in the Riddoch syndrome. *Journal of Neurology*.

9. Rasche S* & **Beyh A***, Paolini M, Zeki S (2023). The neural determinants of abstract beauty. *European Journal of Neuroscience*.

8. O' Brien S, Sethi A, Blair J, Viding E, **Beyh A**, … Craig MC (2023). Rapid white matter changes in children with conduct problems during a parenting intervention. *Translational Psychiatry*.

7. **Beyh A**, Dell'Acqua F, Cancemi D, De Santiago Requejo F, Ffytche D, Catani M (2022). The medial occipital longitudinal tract supports early stage encoding of visuospatial information. *Communications Biology*.

6. Mackes N, Mehta M, **Beyh A**, … Sonuga-Barke E (2022). A prospective study of the impact of severe childhood deprivation on brain white matter in adult adoptees: widespread localized reductions in volume but unaffected microstructural organization. *eNeuro*.

5. Rajashekar D & Lavrador JP, Ghimire P, Keeble H, Harris L, Pereira N, Patel S, **Beyh A**, Gullan R, Ashkan K, Bhangoo R, Vergani F (2022). Simultaneous motor and visual intraoperative neuromonitoring in asleep parietal lobe surgery: dual-strip technique. *Journal of Personalized Medicine*.

4. Brogna C, Lavrador JP, Kandeel HS, **Beyh A**, Ribas EC, Vergani F, Tolias CM (2021). Medial-tonsillar telovelar approach for resection of a superior medullary velum cerebral cavernous malformation: anatomical and tractography study of the surgical approach and functional implications. *Acta Neurochirurgica*.

3. Mirchandani AS, **Beyh A**, Lavrador JP, Howells H, Dell'Acqua F, Vergani F (2021). Altered corticospinal microstructure and motor cortex excitability in gliomas: an advanced tractography and transcranial magnetic stimulation study. *Journal of Neurosurgery*.

2. Howells H, Thiebaut de Schotten M, Dell'Acqua F, **Beyh A**, Zappalà G, Leslie A, Simmons A, Murphy DG, Catani M (2018). Frontoparietal tracts linked to lateralized hand preference and manual specialization. *Cerebral Cortex*.

1. Catani M, Robertsson N, **Beyh A**, … Dell'Acqua F (2017). Short parietal lobe connections of the human and monkey brain. *Cortex*.

Published Book Chapters

2. **Beyh A**, Ohlerth AK, Forkel SJ (2025). Harnessing advanced tractography in neurosurgical practice. In: Krieg SM & Picht T (Eds.), *Navigated Transcranial Magnetic Stimulation in Neurosurgery*.

1. Forkel SJ, Bortolami C, Dulyan L, Barrett R, **Beyh A** (2025). Dissecting white matter pathways: a neuroanatomical approach. In: Dell'Acqua F, Descoteaux M, Leemans A (Eds.), *Handbook of Diffusion MRI Tractography*.

Preprints & Articles Under Review

Ozker M, Giglio L, **Beyh A**, Forkel S, Hagoort P (2025). Individual differences in speech monitoring: functional and structural correlates of delayed auditory feedback. *BioRxiv*.

Dell'Acqua F* & **Beyh A***, … Catani M (2025). MegaTrack: a framework for the anatomically accurate and time-efficient virtual dissection and analysis of large-scale tractography datasets. *BioRxiv*.

Matsulevits A, Alvez P, Atzori M, **Beyh A**, … Thiebaut de Schotten M (2025). A global effort to benchmark predictive models and reveal mechanistic diversity in long-term stroke outcomes. *BioRxiv*.

Citro S, Dawson M, **Beyh A**, … Catani M. Structural evolution of the frontal aslant tract: implications for primate vocalization and human speech. *Under review at Nature Communications.*

Works in Progress

Beyh A, Kim JZ, Larson B, Parkes L. Using anatomo-functional constraints to train biologically informed recurrent neural networks.

Kapteijns B, **Beyh A**, … Forkel S. Specific white matter pathways of reading and mathematics in atypical learners: a transdiagnostic perspective on early learning and attention.

Dulyan L, Bortolami C, Guzmán Chacón E, **Beyh A**, Thiebaut de Schotten M, Forkel S. Manifold lateralisation and variability in the language connectome at 7 T.

GRANTS

1K99MH145984, K99/R00 NIH Pathway to Independence Award — 2026–2031
National Institute of Mental Health
Title: Computational modeling of the brain structure-function links underlying hallucinations in psychosis and Parkinson's disease
Role: PI
Under review

FELLOWSHIPS, AWARDS, & SCHOLARSHIPS

Fellowships

Fellow, The Philadelphia Symposium for Postdoctoral Initiatives in Neuroscience Excellence — 2025

Awards & Honors

Best Poster Award, Brain Health Institute, Rutgers University — 2025
Best Abstract Award, European Workshop on Cognitive Neuropsychology — 2018
Finalist, Wellcome Image Awards, The Wellcome Trust — 2017
The Henry Mc Ilwain Best Poster Award, King's College London — 2015

Scholarships & Travel Support

Travel Grant, Brain Health Institute, Rutgers University — 2025
Scholarship, CIFAR Neuroscience of Consciousness Winter School — 2022
Travel Grant, Centre for Doctoral Studies, King's College London — 2019
Travel Grant, Guarantors of Brain — 2019

TEACHING & SPEAKING ENGAGEMENTS

Guest Lectures

Networks of the visual system (King's College London) — 2025–2026
Introduction to neuroimaging (University College London) — 2020–2022
Occipital lobe anatomy & function (The London Neuromonitoring and Brain Mapping Course) — 2022
Neuroanatomy for neuroimagers (King's College London) — 2020

Introduction to diffusion MRI (King's College London) 2019
Attendee score: 4.7/5

Networks of the visual system (King's College London) 2016–2018

Invited Talks & Research Seminars

Bridging neuroscience & AI with biologically constrained neural networks (University of Pennsylvania) 2025

From diffusion MRI to white matter pathways (Rutgers University) 2025
Attendee score: 4.7/5

bioRNNs: toward more biologically plausible neural network models of the brain (Rutgers University) 2025

Medial occipito-temporal networks (University of Bordeaux) 2023

Callosal connections and visual motion processing (Neuroccino, Clinical Neuroanatomy Seminars) 2023

Occipital lobe anatomy & function (The London Neuromonitoring and Brain Mapping Course) 2022

GPS and spatial memory (Neuroccino, Clinical Neuroanatomy Seminars) 2022

Workshops & Technical Instruction

Introduction to diffusion MRI preprocessing, NEURAL Workshop (virtual) 2025
Attendee score: 5/5

Diffusion MRI, Neuroanatomy, and Tractography Workshop (University of Auckland) 2023–2024
Attendee score: 4.6/5

Diffusion MRI preprocessing, Symposium on Diffusion Imaging (UCLouvain) 2018

Neuroanatomy and Tractography Workshop, NatBrainLab (King's College London) 2016–2018

ACADEMIC SERVICE & LEADERSHIP

Editorial Roles

Consulting Editor, Cortex 2025–

Peer Review — Scientific Journals

Brain Communications · Brain Structure & Function · Communications Biology · Cortex · eLife · Human Brain Mapping · Nature Communications · Network Neuroscience · NeuroImage · Neurosurgical Review · PNAS · Scientific Reports

Peer Review — Funding Agencies

Brain Research UK · French National Research Agency

Committee Roles

Secretary, Neuroscience Alliance 2024–

Organizer, Neural Networks Journal Club, Rutgers University 2023–

Member, Organizing Committee, Neuroanatomy and Tractography Workshop, NatBrainLab 2016–2021

Member, Organizing Committee, Clinical Neuroanatomy Seminars 2015–2018

Member, Organizing Committee, London BrainHack 2017

Outreach & Community Engagement

Scientific Consultant, National Art Pass Campaign, Art Fund UK	2023
Panelist, Discover Careers in Clinical Practice, King's College London	2021
Panelist, IoPPN Alumni Career Event, King's College London	2015
Member, Organizing Committee, Autism Awareness Campaign, American University of Beirut	2014

CONFERENCE PRESENTATIONS

Contributed Talks

American Physical Society (Anaheim, CA, USA)	2025
CIFAR Neuroscience of Consciousness Winter School (Cancun, Mexico)	2022
International Society for Magnetic Resonance in Medicine, Diffusion Study Group (virtual)	2021
Gray's Neuroanatomy Conference (Siracusa, Sicily, Italy)	2018
European Workshop on Cognitive Neuropsychology (Bressanone, Italy)	2018

Posters

Rutgers Brain Health Institute Annual Symposium (New Brunswick, NJ, USA)	2025
Organization for Human Brain Mapping (Rome, Italy)	2019
Organization for Human Brain Mapping (Vancouver, Canada)	2017
Organization for Human Brain Mapping (Geneva, Switzerland)	2016
Neuroimaging and Experimental Medicine Advances (London, UK)	2015

PROFESSIONAL DEVELOPMENT

Deep Learning Course, Neuromatch Academy	2024
The Human Connectome Project Course	2018
Preparing to Teach in Higher Education, King's College London	2016
The FSL Course, Oxford Centre for Functional MRI of the Brain	2016